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Integrated speaker and mobile terminal

Abstract

The present application provides an integrated speaker and a mobile terminal. The integrated speaker includes a housing, a heat dissipation assembly configured to drive air and dissipate heat from the mobile terminal, and a sounding assembly configured to improve the sound performance of the mobile terminal, including a sounding driver and a cover plate. A rear cavity is formed between the cover plate and the sounding driver, and a front cavity is formed between the sounding driver and the housing. The housing is provided with a sound outlet hole connected to the front cavity, a partition plate a first and second through-holes. The sound of the sounding assembly is emitted from the sound outlet hole. The rear and front cavities are configured to improve the audio experience effect of the mobile terminal, and the housing plays the role of protecting the heat dissipation assembly and the sounding assembly.

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Background/Summary

TECHNICAL FIELD

(1) The present application relates to the field of speaker technologies, in particular to an integrated speaker and a mobile terminal.

BACKGROUND

(2) With the advancement of technology, mobile terminal speakers have evolved into multiple-speaker configurations as an integral part of mobile devices.

(3) Typically, the multi-speaker form of the mobile terminal in the related art is usually set up with speaker modules at the top and bottom of the mobile terminal respectively to form a speaker acoustic combination. However, the acoustic performance of solutions for mobile terminal speaker setups in the related art is often limited. Although adding a subwoofer module can improve acoustic performance, it conflicts with the positioning of a peripheral radiator in the mobile terminal, and it is impossible to add the peripheral radiator after adding the subwoofer module.

(4) Therefore, it is necessary to provide an integrated speaker and a mobile terminal, which can not only enhance the audio experience effect of the mobile terminal, but also realize the heat dissipation of the mobile terminal.

SUMMARY

(5) An objective of the present application is to provide an integrated speaker and a mobile terminal, which can not only enhance the audio experience effect of the mobile terminal, but also realize the heat dissipation of the mobile terminal.

(6) The technical solutions of the present application are as follows: an integrated speaker, including a housing with an accommodation cavity; a heat dissipation assembly provided in the accommodation cavity; a sounding assembly provided in the accommodation cavity and spaced from the heat dissipation assembly, including a sounding driver provided in the housing; and a cover plate at least partially attached to the housing, wherein a rear cavity is formed between the cover plate and the sounding driver, and a front cavity is formed between the sounding driver and the housing; wherein the housing is provided with a sound outlet hole connected to the front cavity, and a partition plate separating the accommodation cavity into a first cavity and a second cavity; a side wall of the housing is provided with a first through-hole connected to the first cavity, and a second through-hole connected to the second cavity; the heat dissipation assembly is provided in the first cavity; the heat dissipation assembly is configured to drive air from the second cavity to the first cavity, or, the heat dissipation assembly is configured to drive air from the first cavity to the second cavity.

(7) In one embodiment, the heat dissipation assembly includes a heat conducting plate capped at an opening of the housing, a heat dissipation member provided on the heat conducting plate and located in the first cavity, and a fan assembly provided on the heat dissipation member; wherein the heat dissipation member includes a base plate attached to the heat conducting plate and a heat dissipation structure attached to the base plate and surrounding an outer side of the fan assembly; a plurality of gaps are formed in the heat dissipation structure; and the fan assembly is configured to drive air from the second cavity to the first cavity through the plurality of gaps.

(8) In one embodiment, the heat dissipation structure includes at least two first heat dissipation plates and at least two second heat dissipation plates, wherein the at least two first heat dissipation plates and at least two second heat dissipation plates are spaced apart and around the outer side of the fan assembly, an end of each first heat dissipation plate away from the partition plate and an end of each second heat dissipation plate away from the base plate are abutted against the partition plate.

(9) In one embodiment, the first heat dissipation plates are staggered with the second heat dissipation plates, and the second heat dissipation plates are provided around an outer side of each first heat dissipation plate.

(10) In one embodiment, the partition plate is provided with a third through-hole opposite to a position of the fan assembly, wherein the third through-hole is connected to the first cavity and the second cavity.

(11) In one embodiment, the partition plate further includes a flange extending from an edge of the third through-hole towards the heat dissipation assembly, wherein an outer side of the flange is

abuted against an inner side of each of the first heat dissipation plates.

(12) In one embodiment, the cover plate is provided with a convex portion projecting towards the heat dissipation assembly and corresponding to a position of the third through-hole, and a fitting portion extending outwardly from an outer edge of the convex portion and fixed to the housing, wherein the convex portion is extended at least partially into the third through-hole.

(13) In one embodiment, a heat conducting member is provided between the base plate and the housing, and the heat conducting member is made of heat conductive material.

(14) In one embodiment, a first limiting slot is provided on an inner side of the housing and a second limiting slot is provided around an outer side of the first limiting slot; the sounding driver is embedded in the first limiting slot, and the cover plate is embedded in the second limiting slot.

(15) The beneficial effect of the present application is to provide an integrated speaker on the one hand, and a mobile terminal provided with the above-mentioned integrated speaker on the other hand. In the integrated speaker of the present application, the sounding assembly and the heat dissipation assembly are provided in the housing with an accommodation cavity, so that the present application can not only increase the audio effect of the mobile terminal after being attached to the mobile terminal, but also dissipate heat from the mobile terminal. The heat dissipation assembly is configured to dissipate heat from the mobile terminal, and the sounding assembly is configured to improve the sound performance of the mobile terminal. The heat dissipation assembly drives the air through the first cavity and the second cavity to bring out the heat generated during the operation of the mobile terminal, so as to dissipate heat from the mobile terminal. When the sounding assembly is operating, the sound is emitted through the sound outlet hole, and the rear cavity and the front cavity are configured to improve the sound quality of the sounding assembly and improve the audio experience of the mobile terminal. In addition, the housing can also play a role in protecting the heat dissipation assembly and the sounding assembly, and the heat dissipation assembly and the sounding assembly sharing a housing makes the overall structure of the speaker simplified.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a diagram illustrating an overall structure of a mobile terminal with an integrated speaker according to an embodiment of the present application.

(2) FIG. 2 is a sectional view of the integrated speaker along the A-A direction according to an embodiment of the present application.

(3) FIG. 3 is a diagram of a three-dimensional decomposition structure of the integrated speaker according to an embodiment of the present application.

(4) FIG. 4 is a structural diagram of a heat dissipation member in the integrated speaker according to an embodiment of the present application.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(5) The present application is further described below in conjunction with the accompanying drawings and embodiments.

(6) As shown in FIGS. 1-3, this embodiment provides an integrated speaker **1** and a mobile terminal **2**, and the mobile terminal **2** includes a mobile terminal body and the integrated speaker **1** attached to a rear cover of the mobile terminal **2**. The mobile terminal body may be a cell phone, a tablet computer, and other similar terminal devices. The integrated speaker **1** includes a housing **11** with an accommodation cavity and a heat dissipation assembly **12** arranged in the accommodation cavity. The integrated speaker **1** further includes a sounding assembly **13** arranged in the accommodation cavity and spaced apart from the heat dissipation assembly **12**. The sounding assembly **13** includes a sounding driver **132** arranged in the housing **11** and a cover plate **131** at least partially attached to the housing **11**. A rear cavity **112** is formed between the cover plate **131**

and the sounding driver **132**, and a front cavity **111** is formed between the sounding driver **132** and the housing **11**. The housing **11** is provided with a sound outlet hole **113** connected to the front cavity **111**. The housing **11** is further provided with a partition plate **14** separating the accommodation cavity into a first cavity **114** and a second cavity **115**, and a side wall of the housing **11** is provided with a first through-hole **116** connected to the first cavity **114** and a second through-hole **117** connected to the second cavity **115**. The heat dissipation assembly **12** is arranged in the first cavity **114**, and the heat dissipation assembly **12** is configured to drive air from the second cavity **115** to the first cavity **114**. Alternatively, the heat dissipation assembly **12** is configured to drive air from the first cavity **114** to the second cavity **115**.

(7) In this embodiment, an integrated speaker **1** is provided on the one hand, and a mobile terminal **2** is provided with the above-mentioned integrated speaker **1** on the other hand. In the integrated speaker **1** of the present application, the sounding assembly **13** and the heat dissipation assembly **12** are arranged in the housing **11** with the accommodation cavity, so that the present application can both increase the audio effect of the mobile terminal and dissipate heat for the mobile terminal **2** after being attached to the mobile terminal **2**. The heat dissipation assembly **12** is configured to dissipate heat from the mobile terminal **2**, and the sounding assembly **13** is configured to improve the sound performance of the mobile terminal **2**. The heat dissipation assembly **12** drives air through the first cavity **114** and the second cavity **115** to bring out the heat generated during the operation of the mobile terminal **2**, so as to dissipate heat from the mobile terminal **2**. When the sounding assembly **13** is operating, the sound is emitted through the sound outlet hole **113**, and the rear cavity **112** and the front cavity **111** are configured to improve the sound quality of the sounding assembly **13** and improve the audio experience effect of the mobile terminal **2**. In addition, the housing **11** can also play the role of protecting the heat dissipation assembly **12** and the sounding assembly **13**, and the heat dissipation assembly **12** and the sounding assembly **13** sharing the housing **11** makes the overall structure of the integrated speaker **1** simplified. There are a plurality of first through-holes **116** and a plurality of second through-holes **117**, which are arranged around the side wall of the housing **11**. Each of the first through-holes **116** is actually a hole formed by a notch on the housing **11** near the opening and the heat dissipation assembly **12**. There are a plurality of sound outlet holes **113** provided on the top of the housing **11**, including a center hole arranged in a center of the top of the housing **11** and auxiliary holes arranged around the center hole. The sound outlet holes **113** are provided to improve the sound effect of the integrated speaker **1**, and the solution in this embodiment is the optimal design approach. In other embodiments, the heat dissipation assembly **12** may drive air from the first cavity **114** to the second cavity **115**. In other embodiments, it is also possible to provide a single first through-hole **116**, a second through-hole **117**, or a sound outlet hole **113**, and the number of first through-hole **116**, second through-hole **117**, or sound outlet hole **113** is not limited in the present application. In other embodiments, the first through-hole **116** may also be a hole near the opening of the housing **11**.

(8) As shown in FIGS. 2-3, the heat dissipation assembly **12** includes a heat conducting plate **121** capped at the opening of the housing **11**, a heat dissipation member **122** arranged on the heat conducting plate **121** and located in the first cavity **114**, and a fan assembly **123** arranged on the heat dissipation member **122**. The heat dissipation member **122** includes a base plate **1222** attached to the heat conducting plate **121** and a heat dissipation structure **1221** attached to the base plate **1222** and surrounding an outer side of the fan assembly **123**. A plurality of gaps within the heat dissipation structure **1221**. The fan assembly **123** is configured to drive air from the second cavity **115** to the first cavity **114** through the gaps.

(9) In this embodiment, the heat conducting plate **121** is capped at the opening of the housing **11**. The heat dissipation assembly **12** is attached to the heat conductor **15** and is located in the accommodation cavity of the housing **11**. A side of the heat conducting plate **121** that is back from the housing **11** is attached to the mobile terminal **2**. The heat generated by the mobile terminal **2** during operation is conducted to the heat dissipation member **122** through the heat conducting plate

121, and the fan assembly **123** drives the air in the first cavity **114** and the second cavity **115** to flow when the fan assembly **123** operates, and the heat generated by the mobile terminal **2** is brought out through the flowing air, thereby achieving heat dissipation of the mobile terminal **2**. In this embodiment, the first cavity **114** and the first through-hole **116** form an air outlet channel, and the second cavity **115** and the second through-hole **117** form an air inlet channel. In other embodiments, it may also be that the first cavity **114** and the first through-hole **116** form an air inlet channel, and the second cavity **115** and the second through-hole **117** form an air outlet channel.

(10) As shown in FIGS. 3-4, the heat dissipation structure **1221** includes at least two first heat dissipation plates **12211** and at least two second heat dissipation plates **12212**. The at least two first heat dissipation plates **12211** and the at least two second heat dissipation plates **12212** are spaced apart and around an outer side of the fan assembly **123**. An end of each first heat dissipation member **12211** away from the base plate **1222** and an end of each second heat dissipation member **12212** away from the base plate **1222** are abutted against the partition plate **14**, respectively.

(11) In this embodiment, a plurality of first heat dissipation plates **12211** and a plurality of second heat dissipation plates **12212** are provided on the outer side of the fan assembly **123**. The first heat dissipation plates **12211** and second heat dissipation plates **12212** are rectangular and extended in the radial direction of the fan assembly **123**. The partition plate **14** is arranged in the accommodation cavity and is abutted against each of the first heat dissipation plates **12211** and second heat dissipation plates **12212**. The air circulation channels are formed between the adjacent first heat dissipation members **12211** and the partition plate **14**, and the air circulation channels are also formed between the adjacent second heat dissipation members **12212** and the partition plate **14**, so that when the fan assembly **123** is in operation, the air circulates through the channels to carry away the heat conducted by the heat conducting plate **121** to the heat dissipation member **122**.

(12) As shown in FIG. 4, the first heat dissipation plates **12211** are staggered with the second heat dissipation plates **12212**, and each second heat dissipation plate **12212** is provided around an outer side of each first heat dissipation plate **12211**. In this embodiment, by providing the staggered first heat dissipation plates **12211** and the second heat dissipation plates **12212**, and arranging the second heat dissipation plate **12212** around the outer side of each first heat dissipation plate **12211**, the air has an increased contact surface with the heat dissipation member **122** after entering the first cavity **114** to absorb more heat, thus improving the heat dissipation effect of the heat dissipation assembly **12**.

(13) As shown in FIGS. 2-3, the partition plate **14** is provided with a third through-hole **141** opposite to the position of the fan assembly **123** and a flange **142** extending from an edge of the third through-hole **141** toward the heat dissipation assembly **12**. The flange **142** is abutted against the first heat dissipation plate **12211**, and the third through-hole **141** is connected to the first cavity **114** and the second cavity **115**.

(14) In this embodiment, the first cavity **114** and the second cavity **115** are connected to the third through-hole **141** to allow air to circulate through the first cavity **114** and the second cavity **115**, so that the air can carry away the heat conducted to the heat dissipation plate.

(15) As shown in FIGS. 2-3, the partition plate **14** is provided with a flange **142** extending from an edge of the third through-hole **141** toward the heat dissipation assembly **12**, and the flange **142** is abutted against the first heat dissipation plate **12211**. In this embodiment, the flange **142** is configured to direct the airflow to the first cavity **114** on the one hand, to facilitate the assembly of the partition plate **14** with the first heat dissipation plates **12211**, and to limit the partition plate on the other hand.

(16) As shown in FIG. 3, the cover plate **131** has a convex portion **1311** projecting toward the heat dissipation assembly **12** and corresponding to the position of the third through-hole **141**, and a fitting portion **1312** extending outwardly from an outer edge of the convex portion **1311** and fixed to the housing **11**. The convex portion **1311** is extended at least partially toward the third through-

hole **141**.

(17) In this embodiment, the housing **11** and a part of the fitting portion **1312** are partly fixed together. The rear cavity **112** of the sounding assembly **13** is formed among the portion of the fitting portion **1312** that is not fitted to the housing **11**, the convex portion **1311**, and the sounding driver **132**, and the rear cavity **112** can serve to improve the sound quality of the sounding assembly **13** when the sounding assembly **13** is in operation. The convex portion **1331** is partially inserted into the third hole **141** to guide the air to the first cavity **114** on the one hand, and to increase the volume of the rear cavity **112** on the other hand, so that the rear cavity **112** can better improve the sound quality of the sounding assembly **13**.

(18) As shown in FIGS. 2-3, a heat conducting member **15** is further provided between the base plate **1222** and the housing **11**, which is made of heat conductive material. In this embodiment, the heat conducting member **15** is configured to quickly conduct the heat generated by the mobile terminal **2** to the heat dissipation member **122**, and a fan base **1231** and fan blades **1232** of the fan assembly **123** are rotatably connected. The fan blades **1232** continuously rotate to make the airflow quickly take away heat, thereby reducing the temperature of the mobile terminal **2** itself.

(19) As shown in FIG. 2, an inner side of the housing **11** is provided with a first limiting slot **118**, and a second limiting slot **119** surrounding an outer side of the first limiting slot **118**. The sounding driver **132** is embedded in the first limiting slot **118** and the cover plate **131** is embedded in the second limiting slot **119**. In this embodiment, the first limiting slot **118** is configured to limit the sounding driver **132**, and the second limiting slot **119** is configured to limit the cover plate **131**, thereby preventing displacement of the sounding driver **132** and the cover plate **131**, which may affect the overall use of the integrated speaker **1**.

(20) Described above are only embodiments of the present application, and it should be noted that, for those of ordinary skill in the art, improvements made without departing from the conceptual essence of the present application shall fall within the protection scope of the present application.

Claims

1. An integrated speaker, comprising: a housing with an accommodation cavity; a heat dissipation assembly provided in the accommodation cavity; a sounding assembly provided in the accommodation cavity and spaced from the heat dissipation assembly, comprising: a sounding driver provided in the housing; and a cover plate at least partially attached to the housing, wherein a rear cavity is formed between the cover plate and the sounding driver, and a front cavity is formed between the sounding driver and the housing; wherein the housing is provided with a sound outlet hole connected to the front cavity, and a partition plate separating the accommodation cavity into a first cavity and a second cavity; a side wall of the housing is provided with a first through-hole connected to the first cavity, and a second through-hole connected to the second cavity; the heat dissipation assembly is provided in the first cavity; the heat dissipation assembly is configured to drive air from the second cavity to the first cavity, or, the heat dissipation assembly is configured to drive air from the first cavity to the second cavity; the heat dissipation assembly comprises a heat conducting plate capped at an opening of the housing, a heat dissipation member provided on the heat conducting plate and located in the first cavity, and a fan assembly provided on the heat dissipation member; wherein the heat dissipation member comprises a base plate attached to the heat conducting plate and a heat dissipation structure attached to the base plate and surrounding an outer side of the fan assembly; a plurality of gaps are formed in the heat dissipation structure; and the fan assembly is configured to drive air from the second cavity to the first cavity through the plurality of gaps.

2. The integrated speaker of claim 1, wherein the heat dissipation structure comprises at least two first heat dissipation plates and at least two second heat dissipation plates, wherein the at least two first heat dissipation plates and at least two second heat dissipation plates are spaced apart and

around the outer side of the fan assembly; an end of each first heat dissipation plate away from the partition plate and an end of each second heat dissipation plate away from the base plate are abutted against the partition plate.

3. The integrated speaker of claim 2, wherein the first heat dissipation plates are staggered with the second heat dissipation plates, and the second heat dissipation plates are provided around an outer side of each first heat dissipation plate.

4. The integrated loudspeaker of claim 2, wherein the partition plate is provided with a third through-hole opposite to a position of the fan assembly, wherein the third through-hole is connected to the first cavity and the second cavity.

5. The integrated speaker of claim 4, wherein the partition plate further comprises a flange extending from an edge of the third through-hole towards the heat dissipation assembly, wherein an outer side of the flange is abutted against an inner side of each of the first heat dissipation plates.

6. The integrated speaker of claim 4, wherein the cover plate is provided with a convex portion projecting towards the heat dissipation assembly and corresponding to a position of the third through-hole, and a fitting portion extending outwardly from an outer edge of the convex portion and fixed to the housing, wherein the convex portion is extended at least partially into the third through-hole.

7. The integrated speaker of claim 1, wherein a heat conducting member is provided between the base plate and the housing, and the heat conducting member is made of heat conductive material.

8. The integrated speaker of claim 1, wherein a first limiting slot is provided on an inner side of the housing and a second limiting slot is provided around an outer side of the first limiting slot; the sounding driver is embedded in the first limiting slot, and the cover plate is embedded in the second limiting slot.

9. A mobile terminal, comprising the integrated speaker of claim 1, wherein the integrated speaker is attached to a rear cover of the mobile terminal.
